Probabilistic CKY Parsing

NLP II Recitation

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Today

1. Quick review of CKY and PCFGs
2. Probabilistic CKY: what changes?
3. Discussion
4. Practice problems

Review: Why CNF?

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This guarantees: every non-leaf node has **exactly 2 children**.

So we can always split a span into exactly two parts.

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$$P(T) = \prod_{\text{rule } r \text{ used in } T} P(r)$$

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What assumption does this make? **Independence** of rule applications.

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PCFGs let us pick the **most likely** parse:

$$T^* = \arg\max_T P(T \mid \text{sentence})$$

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What do we need to track instead for probabilistic parsing?

Probabilistic CKY: $\pi[i, j, A]$ = **probability** of best subtree rooted at A spanning $[i, j]$

Also need: **backpointers** to reconstruct the tree

$$\text{bp}[i, j, A] = (k, B, C) \text{ that achieved the max}$$

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Why max? We want the single best parse (Viterbi-style).

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Benefits?

- Numerical stability
- Addition instead of multiplication

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Is this different from regular CKY? **No!** Same complexity.

Discussion: Viterbi vs. Inside

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When would you want this instead of the best parse?

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What are the pros/cons of each?

Practice 1: Fill the Chart

Grammar:

$$S \to \text{NP VP } [1.0] \quad \text{NP} \to \text{Det N } [1.0] \quad \text{VP} \to \text{V NP } [1.0]$$
$$\text{Det} \to \text{“the”} \ [1.0] \quad \text{N} \to \text{“cat”} \ [0.6] \quad \text{N} \to \text{“dog”} \ [0.4]$$
$$\text{V} \to \text{“chased”} \ [1.0]$$

Sentence: “the cat chased the dog”

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Sentence: "the cat chased the dog"

	the	*cat*	*chased*	*the*	*dog*
	Det: 1.0 [0,1]				
		N: 0.6 [1,2]			
			V: 1.0 [2,3]		
				Det: 1.0 [3,4]	
					N: 0.4 [4,5]

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Work in pairs: fill in the chart.

Practice 1: Solution

	the	*cat*	*chased*	*the*	*dog*
	Det: 1.0 [0,1]	**NP: 0.6** [0,2]	– [0,3]	– [0,4]	**S: 0.24** ← [0,5]
		N: 0.6 [1,2]	– [1,3]	– [1,4]	– [1,5]
			V: 1.0 [2,3]	– [2,4]	**VP: 0.4** [2,5]
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$$\pi[0, 2, \text{NP}] = 1.0 \times 1.0 \times 0.6 = 0.6$$
$$\pi[3, 5, \text{NP}] = 1.0 \times 1.0 \times 0.4 = 0.4$$
$$\pi[2, 5, \text{VP}] = 1.0 \times 1.0 \times 0.4 = 0.4$$
$$\pi[0, 5, S] = 1.0 \times 0.6 \times 0.4 = \mathbf{0.24}$$

Practice 2: Ambiguity

Added rules:

$$\begin{array}{ll} \text{NP} \rightarrow \text{NP PP } [0.3] & \text{NP} \rightarrow \text{Det N } [0.7] \\ \text{VP} \rightarrow \text{VP PP } [0.4] & \text{VP} \rightarrow \text{V NP } [0.6] \\ \text{PP} \rightarrow \text{P NP } [1.0] & \text{P} \rightarrow \text{“with”} \ [1.0] \\ \text{V} \rightarrow \text{saw } [1.0] & \text{N} \rightarrow \text{man } [1.0] \\ \text{N} \rightarrow \text{telescope } [1.0] & \text{Det} \rightarrow \text{the } [1.0] \end{array}$$

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Calculate the probability of both parses. Which does the PCFG prefer?

Hint: The only difference is which rule combines the PP.

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Parse 1: PP attaches to NP ("the man with the telescope")

$$\begin{aligned} P &= P(\text{VP} \rightarrow \text{V NP}) \times P(\text{NP} \rightarrow \text{NP PP}) \times \ldots \\ &= 0.6 \times 0.3 \times (\text{other terms}) = 0.18 \times (\text{other}) \end{aligned}$$

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It's deterministic—just a structural transformation.

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What's the trade-off?

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Garden path sentence—reduced relative clause. The grammar likely lacks the rule or it has near-zero probability from training data.

Summary

- Probabilistic CKY = CKY + probabilities
- Track max probability per cell (Viterbi)
- Use log probs to avoid underflow
- Same $O(n^3)$ complexity
- Backpointers to recover tree
- Limitations: independence assumption, lexical sensitivity

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Next time: Neural constituency parsing (how do we move beyond PCFGs?)

Questions?

Email: timothyobiso@brandeis.edu

Office Hours:
Tuesday 1:00-2:00 pm
Wednesday 2:00-3:00 pm